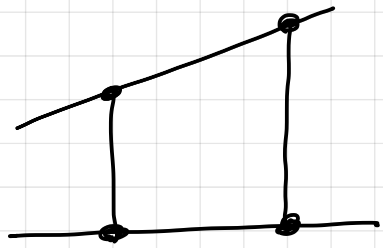
$\boxed{1.7.15}$ One can present the same kind of examples for the ring $R = k[x]$ instead of $R = \mathbb{Z}$. Polynomials have the space attached to them, on which you can evaluate them. In the case of $k[x]$, you can evaluate them on k.

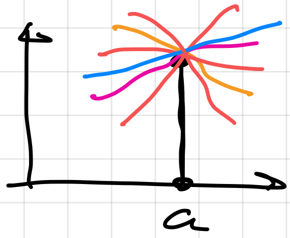


$\boxed{1.7.16}$ The CRT and the interpolation. Let's consider the ring $R = k[x]$ with $k = \mathbb{R}$ (for concreteness).

The ideal $(x-a)\mathbb{R}[x]$ for $a \in \mathbb{R}$ is an ideal of all polynomials that are equal to 0 at a.
Hence, $\mathbb{R}[x]/(x-a)\mathbb{R}[x]$ consists of cosets where for each given coset, the polynomials have the same value in a.

$f \equiv h \mod x-a \iff f-h$ is divisible by $x-a$
$f(a) = h(a)$.

That means $\mathbb{R}[x]/_{(x-a)\mathbb{R}[x]} \cong \mathbb{R}$



via the evaluation $[f]_{x-a} \longmapsto f(a)$.

So, the canonical homomorphism

$$\mathbb{R}[x] \longrightarrow \mathbb{R}[x]/_{(x-a)\mathbb{R}[x]}$$ corresponds

to the evaluation of $f \in \mathbb{R}[x]$ in the given a.

Let's take a more complicated polynomial, which choose one that factorizes into linear factors, say $g(x) = x\cdot(x-1)\cdot(x-2)$.

It gives rise to the ring

$\mathbb{R}[x]/_{g(x)\mathbb{R}[x]}$, $f \equiv h \mod g \iff$

$f - h$ is divisible by $g \iff$

$$f(x) - h(x) = \underbrace{g(x)}_{x\cdot(x-1)\cdot(x-2)} \cdot q(x) \quad \text{for some } q \in \mathbb{R}[x]$$

$$\Longrightarrow \quad \begin{aligned} f(0) - h(0) &= 0 \\ f(1) - h(1) &= 0 \\ f(2) - h(2) &= 0 \end{aligned}$$

The converse is also true.

So a coset $[f]_g = [f]_{x(x-1)(x-2)}$ is completely determined by the triple of values

$(f(0), f(1), f(2))$. So, we see that our ring $\mathbb{R}[x]/g(x)\mathbb{R}[x]$ is isomorphic to the ring $\mathbb{R}\times\mathbb{R}\times\mathbb{R}$ via the isomorphism

$$[f]_g \longmapsto (f(0), f(1), f(2)).$$

Inverting this isomorphism is (essentially) solving the interpolation problem for the nodes 0, 1 and 2.

All this can be interpreted in terms of the CRT.

$$x\,\mathbb{R}[x] \cap (x-1)\mathbb{R}[x] \cap (x-2)\mathbb{R}[x] = g(x)\,\mathbb{R}[x]$$

coprime coprime coprime.

By the CRT

$$\underbrace{\mathbb{R}[x]/g(x)\mathbb{R}[x]}_{\substack{\Updownarrow \\ \text{corresponds} \\ \text{to evaluation} \\ \text{on } \{0,1,2\}}} \cong \underbrace{\underbrace{\mathbb{R}[x]/x\mathbb{R}[x]}_{\substack{\Updownarrow \\ \text{corresponds to} \\ \text{evaluation} \\ \text{in } 0}} \times \underbrace{\mathbb{R}[x]/(x-1)\mathbb{R}[x]}_{\substack{\Updownarrow \\ \text{corresponds to} \\ \text{evaluation} \\ \text{in } 1}} \times \underbrace{\mathbb{R}[x]/(x-2)\mathbb{R}[x]}_{\substack{\Updownarrow \\ \text{corresponds} \\ \text{to} \\ \text{evaluation} \\ \text{in } 2}}}_{\cong\, \mathbb{R}\times\mathbb{R}\times\mathbb{R}}$$

This explains the relation of the CRT to the interpolation.

1.8 Units and inversion in rings

1.8.1 Definition. Let R be a non-zero ring. By $R^\times$ we denote the set of all elements $a \in R$ that have a multiplicative inverse $a^{-1} \in R$, which is an element satisfying $a \cdot a^{-1} = a^{-1} \cdot a = 1$. Another notation is: R^*. Elements of $R^\times$ are called units.

1.8.2 $(R^\times, \cdot)$ is an Abelian group. Not hard to see (exercise).

1.8.3 Examples:

(1) $k[x_1, \dots, x_n]^\times = k \setminus \{0\}$ (k field)

(2) $\mathbb{Z}^\times = \{1, -1\}$

(3) $k^\times = k \setminus \{0\}$ (k field)

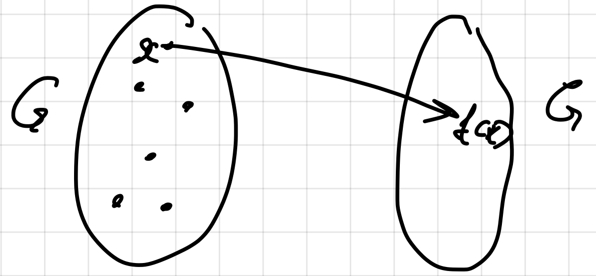
1.8.4 An element r in a ring R is called irreducible if $r \notin R^\times$ and for every factorization $r = a \cdot b$ with $a, b \in R$, one has $a \in R^\times$ or $b \in R^\times$.

For example: in $\mathbb{Z}$, irreducible elements are $\pm$ prime numbers.
For $k[x]$, this corresponds to the notion of irreducible polynomials as we had.

1.8.5 Fields are those unitary commutative rings R in which $R^\times = R \setminus \{0\}$.

1.8.6 In a finite Abelian group $(G, \cdot)$ one has $a^{|G|} = 1$ for every $a \in G$ (the size of the group, $|G|$, is usually called the order of the group).

Proof: Consider the map $f(x) = ax$, $f: G \to G$. It's bijective, because $f^{-1}(x) = a^{-1} \cdot x$ $\left(f(f^{-1}(x)) = x,\ f^{-1}(f(x)) = x \right)$



$\Rightarrow$

$$\prod_{x \in G} x = \prod_{x \in G} f(x) \Rightarrow$$

$$\prod_{x \in G} x = \prod_{x \in G} (a \cdot x) \Rightarrow$$

$$\underbrace{\prod_{x \in G} x}_{= u} = \underbrace{\left(\prod_{x \in G} a\right)}_{a^{|G|}} \cdot \underbrace{\left(\prod_{x \in G} x\right)}_{= u} \Rightarrow$$

$$u = a^{|G|} \cdot u \Rightarrow$$

$$u \cdot u^{-1} = a^{|G|} \cdot u \cdot u^{-1} \Rightarrow$$

$$1 = a^{|G|}.$$

$\square$

1.8.7 **Remark.** 1.8.6 is also true for arbitrary finite groups (not necessarily Abelian). This can be derived from the Lagrange theorem (basic result in group theory).

1.8.8 As a consequence of 1.8.6 we have the following. If we have a finite ring R, then $(R^\times, \cdot)$ is a finite Abelian group so that

$$a^{|R^\times|} = 1 \text{ holds for all } a \in R^\times.$$

This can also be phrased as

$$a^{|R^\times|+1} = a \quad \text{and}$$

$$a^{|R^\times|-1} = a^{-1}.$$

1.8.9 As another consequence of 1.8.6 we have this: if k is a finite field, then $a^{|k|-1} = 1$ holds for every $a \in k \setminus \{0\}$.

This is equivalent to

$$a^{|k|} = a \text{ for every } a \in k.$$

1.8.10 Example for 1.8.8.

$$R = \mathbb{Z}/6 = \{[0]_6, [1]_6, [2]_6, [3]_6, [4]_6, [5]_6\} \Rightarrow$$

$$R^\times = \{[1]_6, [5]_6\}$$

$$[1]_6^2 = [1]_6$$

$$[5]_6^2 = [1]_6$$

Another example: $R = \mathbb{Z}/15$, for it we have $|R^\times| = 8$. We did verification of $a^8 = 1$ for every $a \in R^\times$ in SageMath. (We could have done it in pure Python as well.).

1.8.11 Euler Totient Function. We consider the ring $\mathbb{Z}/n\mathbb{Z}$ with $n \in \mathbb{Z}_{\geq 2}$. The above discussion tells us that $a^{|(\mathbb{Z}/n\mathbb{Z})^\times|} = 1$ for every $a \in \mathbb{Z}/n\mathbb{Z}$. So, we are interested in the number of units in $\mathbb{Z}/n\mathbb{Z}$. This is the function $\varphi(n) := |(\mathbb{Z}/n\mathbb{Z})^\times|$ and is called <u>Euler's totient function</u> or just <u>totient function</u> or <u>Euler's φ-function</u>. One defines $\varphi(1) = 1$. So, this is a function $\varphi: \mathbb{N} \to \mathbb{N}$.

$\boxed{1.8.12}$ Units in quotient rings (invertibility in quotient rings).